Week 1 — Threat Landscape Research

Overview

This week introduced the foundational concepts of cybersecurity by examining the modern threat landscape. The focus was on understanding **who** attacks organizations, **why** they do it, and **how** those attacks are categorized — building the mental model needed for all subsequent security work.

Topics Covered

- Defining cybersecurity and the CIA Triad (Confidentiality, Integrity, Availability)
- Threat actor categories: nation-states, organized crime, hacktivists, insider threats
- Attack taxonomy: understanding kill chains and attack patterns
- OSINT (Open Source Intelligence) fundamentals
- Introduction to risk management frameworks

Assignment

Conducted research into a real-world threat actor or threat group, profiling their:

- Known motivations and targets
- Tactics, Techniques, and Procedures (TTPs)
- Notable attacks and impacts
- Potential defensive countermeasures

Key Concepts

Threat Actor Types

- Nation-state actors (APTs) — highly resourced, long-term campaigns
- Cybercriminals — financially motivated ransomware and fraud
- Hacktivists — ideologically driven disruption
- Insider threats — malicious or negligent internal actors

The CIA Triad

- **Confidentiality** — preventing unauthorized data access
- **Integrity** — ensuring data is accurate and unaltered

- **Availability** — keeping systems accessible to authorized users

Tools & Frameworks Referenced

- MITRE ATT&CK Framework
- OSINT techniques (public records, social media, domain lookups)
- NIST Cybersecurity Framework overview